



Social class differences in dietary intake are mediated by the relationship between health and taste: Findings from a cross-sectional and longitudinal study

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ABSTRACT

In a cross-sectional (Study 1) and longitudinal (Study 2) study, the authors examined whether the associations between social class and dietary intake and between social class and overweight are mediated by the relationship between health and taste. In Study 1 ($N = 1,027$), the assumed ("unhealthy = tasty" belief) and the actual (correlation between health and taste ratings of individual meals) health-taste relationship were assessed as mediators of the association between social class and dietary intake and social class and overweight in a German population. The research question was investigated using correlational analyses and mediation models. The results show consistent associations between social class, daily fruit and vegetable intake, and body mass index (BMI). These associations were partially mediated by the assumed and actual health-taste relationship. In Study 2 ($N = 794$), we analysed social class, the assumed health-taste relationship, dietary intake, and BMI at three different time points between 2011 and 2018 in a Dutch population. Social class predicted dietary intake (but not BMI), partially mediated by the assumed health-taste relationship. Our research points to an underlying mechanism of dietary inequality that goes beyond material resources such as income or education, namely the relationship between health and taste of food.

1. Introduction

Unhealthy diets and overweight are more prevalent among people from lower social classes in most high-income countries (Giskes, Avendaño, et al., 2010; McLaren, 2007). Financial constraints, health education, and the availability and accessibility of healthy or unhealthy foods are frequently discussed explanations for this dietary inequality (Darmon & Drewnowski, 2008, 2015; Rao, Afshin, Singh, & Mozaffarian, 2013; Stormacq, van den Broucke, & Wosinski, 2019; Burgoine et al., 2017). However, these factors cannot fully explain socioeconomic differences in dietary intake (Ball et al., 2006) and more research is needed on the underlying mechanisms of these inequalities (Giskes, van Lenthe, et al., 2010).

The literature highlights that people's lay beliefs (i.e., how people think and reason about food) have a substantial and direct impact on obesity (McFerran & Mukhopadhyay, 2013). Therefore, it is important

to examine how food beliefs are associated with social class and whether food beliefs help to understand differences in food preferences between individuals from different social classes. For example, food beliefs could potentially explain Pechey et al.'s (2015) finding that men from a lower social class like fruit less than men from a higher social class, because lower-class individuals may believe that there is a negative relationship between health and taste of food. However, while there are numerous indications that important beliefs are associated with social class (Kraus et al., 2012; Manstead, 2018), research directly focusing on social class differences in food beliefs is scarce and fragmented (for a review, see van der Heijden et al., 2021) and the effects of food beliefs on eating behaviour are not yet fully understood.

To help explain social class differences in healthy eating, the belief that unhealthy food is tasty is of particular importance. Research on this "unhealthy = tasty" belief showed that individuals often believe that there is a trade-off between the healthiness and tastiness of foods, and

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that this belief leads to the consumption of unhealthier foods (Raghunathan et al., 2006). Stronger unhealthy = tasty beliefs are associated with a higher body mass index (BMI; Mai & Hoffmann, 2015), and this association is mediated by less fruit and vegetable intake (Briers et al., 2020). Interestingly, there is considerable variability in the direction and magnitude of unhealthy = tasty beliefs between people (Haasova & Florack, 2019a; Werle et al., 2013). Thus, individual differences in the unhealthy = tasty belief may contribute to differences in food choices associated with social class.

Consumers likely apply the unhealthy = tasty belief to judge the taste of foods explicitly categorized as healthy or unhealthy. Yet, consumers often face food decisions in contexts where a clear distinction between healthy and unhealthy foods is not apparent. In such situations, consumers may use food cues (e.g., ingredients, method of preparation) to form an expectation about both its healthiness and tastiness independently, which may lead to a correlation between health and taste ratings at the level of individual foods. The correlation is likely negative when consumers use the same cues for a positive taste rating and a negative health rating, or vice versa. For instance, consumers might think that fried food in a meal make the meal tastier, but less healthy. In contrast, the correlation is likely positive when consumers use the same cues for both a positive taste and health rating or a negative taste and health rating. For instance, consumers might think that fresh vegetables in a meal make the meal tastier and healthier, whereas old vegetables make a meal less tasty and less healthy.

Importantly, the actual relationship between people's health and taste ratings for a sample of different foods is moderately correlated with their unhealthy = tasty belief (Haasova & Florack, 2019a), and both are predictors of healthy food preferences (Briers et al., 2020). Thus, the unhealthy = tasty belief and the individual use of cues to infer health and taste are likely to be two complementary and interrelated drivers of food choices.

The objective of the present research was to investigate the link between the different aspects of the health-taste relationship and social class to help understand differences in eating behaviour between individuals of low and high social class. In particular, we measured two aspects of the relationship between health and taste, the assumed relationship, expressed as unhealthy = tasty belief and the actual relationship, expressed as correlation between health and taste ratings across different meals. We examined the associations of these aspects of the health-taste relationship with social class, dietary intake, and overweight. We found that a negative relationship between health and taste (both assumed and actual) is more likely for lower-class individuals than for higher-class individuals, which partially mediates the relationship between social class and dietary intake (fruit and vegetable intake) and between social class and overweight (BMI). The present research contributes to the understanding of the mechanisms underlying dietary inequalities between social classes beyond material resources such as income or education.

2. Conceptual background

2.1. The health-taste relationship

Consumers can form taste expectations based on the perceived healthiness of foods, which influences their food choices and dietary habits (e.g., Briers et al., 2020). In their seminal work, Raghunathan et al. (2006) provided US consumers with explicit information about the healthiness of foods. They found evidence that healthy compared with unhealthy foods are perceived as less tasty, less enjoyable to eat, and are less preferred in a choice task. This so-called "unhealthy = tasty" belief is not tied to any specific food product, but rather to the perceived healthiness and its relationship to taste in general (Raghunathan et al., 2006). Researchers suppose that individual differences in the unhealthy = tasty belief influence how consumers perceive and respond to food options that are explicitly categorized as healthy or unhealthy (Briers

et al., 2020; Raghunathan et al., 2006). Hence, many studies have investigated individual differences in the unhealthy = tasty belief by directly asking consumers whether they think unhealthy foods are tastier than healthy foods (e.g., on a scale-based measure, see Cooremans et al., 2017; van der Heijden et al., 2020). We will refer to this belief as the *assumed* health-taste relationship because it reflects consumers' assumptions about how health and taste are related in reality.

The unhealthy = tasty belief is like a rule of thumb that consumers can apply to judge the taste of foods explicitly categorized as healthy or unhealthy. When foods are not explicitly categorized as healthy or unhealthy, consumers may use food cues (e.g., ingredients, method of preparation) to form an expectation about both its healthiness and tastiness independently, which may lead to a correlation between health and taste ratings at the level of individual foods that consumers are not aware of. The same cues might signal to consumers that a food is both tasty and healthy (e.g., fresh ingredients), or tasty, but unhealthy (e.g., sugar content) or they might signal only taste, but not health (e.g., flavour). Because some cues indicate health and taste in the same or opposite directions, health and taste ratings may correlate even in the absence of explicit health labelling. To study such correlations between health and taste ratings, researchers ask consumers to rate the tastiness and healthiness of different foods separately and then compute correlation coefficients between the health and taste ratings for each individual (Haasova & Florack, 2019a, 2019b; Kunz, Haasova, & Florack, 2020; Kunz, Haasova, Rieß, et al., 2020). We will refer to this correlation between health and taste ratings as *actual* health-taste relationship. The health and taste ratings are still subjective, that is, reflecting consumers' expectations. Yet, consumers do not directly perceive any relationship between health and taste, as they rate them separately and independently and thus do not apply an unhealthy = tasty belief. Hence, the resulting correlation between those ratings reflects the actual relationship between (subjective) health and taste in individual foods.

Contrary to the unhealthy = tasty belief, the actual relationship between healthiness and tastiness has been rarely negative, but moderately positive in most studies implying that consumers often use overlapping cues to form health and taste ratings (Kunz, Haasova & Florack, 2020). Hence, the belief about the health-taste relationship is not always congruent with the actual correlations between health and taste ratings on the individual level. Recent research even suggests that consumers can directly state that unhealthy foods are tastier than healthy foods, despite observing environments in which healthy foods are tastier than unhealthy foods (Kunz et al., 2023).

The observed discrepancies between the actual and assumed health taste relationship imply that they are at least partly governed by different processes. Because the assumed health taste relationship is a belief, it is likely to be influenced by social learning (Turnwald et al., 2022) and social cognitive biases (Kunz et al., 2023). The unhealthy = tasty belief corresponds to the palatable portrayal of unhealthy foods in films and TV series (Turnwald et al., 2022), and children may learn from food parenting practices (i.e., rules and own behaviour) that unhealthy foods, but not healthy foods are used as a reward (Fleary & Ettienne, 2019; Puhl & Schwartz, 2003). Moreover, tasty unhealthy food is often frequent in many food environments, while healthy food is often rare and thus, consumers might link tastiness to unhealthiness (Kunz et al., 2023). On the other hand, the actual health-taste relationship may be the result of a positive overlap between cues signalling health and cues signalling tastiness. That is, consumers may use the same cues to infer good health and taste in a food product, and therefore, the correlation may be often positive. Indeed, Kunz, Haasova and Florack (2020) showed that colour saturation indicates a better healthiness and tastiness of individual foods, contributing to a positive health-taste relationship.

While the assumed and actual health-taste relationship are two distinct aspects of the health-taste relationship, they are not completely unrelated either. In studies assessing both aspects, individual differences in the assumed relationship (the belief) and the actual relationship

(correlations between ratings) are moderately correlated (Haasova & Florack, 2019a) and both predict food choice (Briers et al., 2020).

2.2. Social class differences in the health-taste relationship

A basic idea of the present research is that beliefs about the assumed relationship between health and taste, as well as the use of cues contributing to actual correlations between health and taste ratings, may be patterned by social class. Children from lower socioeconomic neighbourhoods engage in more screen time and TV minutes than those from higher socioeconomic neighbourhoods (Carson et al., 2010), increasing the likelihood that lower-class children will be exposed to messages in line with an unhealthy = tasty belief through the media. Parents of lower socioeconomic status are also more likely to provide their children with sweet snacks as rewards (Damen et al., 2019). Lower-class individuals also place greater importance on familiarity in their food choices and are more likely to avoid unfamiliar foods, a behaviour associated with unhealthy dietary intake (Hazley et al., 2022; Konttinen et al., 2013). Therefore, lower-class individuals may be expected to sample a smaller variety of foods, which may affect their recognition and utilization of cues when assessing the healthiness and tastiness of foods. As a result, lower-class individuals may use the same cues differently than higher-class individuals or use different cues altogether. Therefore, we expect lower-class individuals to express a more negative assumed and actual health-taste relationship.

3. The present research

In the current research, we propose that differences in the relationship between health and taste contribute to dietary inequality between social classes in high-income countries. We examined the associations between social class, the relationship between health and taste (assumed and actual), and dietary intake and overweight in a cross-sectional study with a German population and in a longitudinal design with a sample from the Netherlands.

The first aim of this research was to examine the association between social class and the health-taste relationship. In one study, van der Heijden et al. (2020) measured the unhealthy = tasty belief of 37 parent-child dyads from lower socioeconomic backgrounds and found that parents perceive a positive relationship between health and taste, whereas children perceive a negative relationship. However, van der Heijden et al. (2020) did not compare these results with people from higher socioeconomic backgrounds in a larger sample. Further, we distinguish between an assumed and an actual health-taste relationship. Literature implies that there may be two distinct processes underlying the assumed and actual health-taste relationship, as the assumed health-taste relationship is often negative (Raghunathan et al., 2006), whereas the actual relationship is often positive (Haasova & Florack, 2019a). Therefore, the associations with social class and the effects on dietary intake may differ between these two aspects.

Our second aim was to examine whether the health-taste relationship (both assumed and actual) mediate the associations between social class and dietary intake, and between social class and overweight. If the health-taste relationship is socially patterned, dietary inequality may persist even when healthy foods become more affordable and accessible to lower-class individuals. By investigating this psychological mechanism, we aim to shed light on potential ways to address dietary inequality and contribute to the existing literature on the association between social class and dietary intake.

4. Study 1: Cross-sectional study in Germany

Study 1 served two purposes. First, we examined the associations between social class and the assumed and actual health-taste relationship. Second, we investigated whether the assumed and actual health-taste relationship mediate the association between social class and

dietary intake (daily fruit and vegetable intake) and between social class and overweight (BMI).

4.1. Method

4.1.1. Participants

A web-based cross-sectional survey was conducted with the help of a German panel to obtain a sample that matched the German population in terms of age, gender, BMI, and social class. Using G*Power (Faul et al., 2007), we pre-determined a sample size of 780 to detect small-sized correlations ($r = 0.10$) with an alpha level of 0.05 and a statistical power of 0.80. We instructed the panel to recruit 1,000 participants to rule out any power considerations. Participants who did not eat meat (e. g., vegetarians or vegans) or were on a restricted diet at the time of the study were excluded a-priori. We also decided to only recruit participants from larger German cities (more than 100,000 inhabitants), as rural-urban differences in food intake can be expected but are not the focus of this study. We finished with a sample of 1,027 German adults (49% women, 51% men) who met our criteria, with a mean age of 51.24 years ($SD = 14.35$) and a mean BMI of 27.06 ($SD = 6.48$).⁴ A description of the demographic distribution of our sample can be found in Table 1.

Table 1
Demographic distribution of participants in Study 1 (N = 1,027).

Category	Percentage
<i>Gender</i>	
Male	51%
Female	49%
<i>Monthly net income (n = 954)^a</i>	
Below EUR 500	5%
EUR 500 – EUR 999	13%
EUR 1,000 – EUR 1,999	35%
EUR 2,000 – EUR 2,999	27%
EUR 3,000 – EUR 3,999	11%
EUR 4,000 – EUR 4,999	5%
Greater than EUR 5,000	4%
<i>Educational attainment (n = 1,019)^a</i>	
No degree	<1%
General secondary school	9%
Intermediate secondary school	24%
Vocational and grammar school	38%
University degree	28%
<i>Social class^b</i>	
$M_{\text{Social Class}}$	5.67
$SD_{\text{Social Class}}$	1.70
<i>Age</i>	
M_{Age}	51.24
SD_{Age}	14.35
<i>Overweight^c</i>	
BMI < 25	45%
BMI ≥ 25	55%

Note. M = mean, SD = standard deviation.

^a Sample size differs for the monthly net income ($n = 954$) and education ($n = 1,019$) because participants declined to indicate their income category or their highest educational attainment did not fall into one of the categories listed.

^b Social class was measured using the social class ladder (Adler et al., 2000), where 1 represents the bottom of the society and 10 the top of the society.

^c A BMI ≥ 25 is considered overweight for adults according to the European Commission (Eurostat, n.d.).

⁴ Initially, we finished with a sample of $N = 1,031$ but we could not calculate the actual health-taste relationship for four participants because they had no variance in their health or taste ratings. These four participants were excluded from all further analyses.

4.1.2. Measures

To measure social class, participants saw a ladder representing where people stand in German society and selected the rung on which they ranked themselves compared to all other people in Germany (1 = *at the bottom*, 10 = *at the top*; Adler et al., 2000). The social class ladder has been used in previous studies (e.g., Belmi et al., 2020) to measure social class and correlates positively with educational attainment, $r(1019) = 0.31$, $p < .001$, and monthly net income, $r(954) = 0.47$, $p < .001$, in this study.

For the assumed health-taste relationship, we measured the unhealthy = tasty belief with two scales. First, participants indicated whether making a meal healthier would make it less tasty or tastier and whether making a meal tastier would make it healthier or healthier on two slider bars from 1 to 100 (Cronbach's $\alpha = 0.81$). As a second measure of the unhealthy = tasty belief, we used two items developed by Raghunathan et al. (2006) and a third item added by Mai and Hoffmann (2015). Participants indicated their agreement on a 9-point scale (1 = *strongly disagree*, 9 = *strongly agree*). The three items are: "Things that are good for me rarely taste good", "There is no way to make food healthier without sacrificing taste", and "Healthy food is usually less tasty" (Cronbach's $\alpha = 0.81$). We recoded the scale so that high scores indicated a positive relationship between health and taste and low scores indicated a negative relationship. We decided to use two scales because although the original scale by Raghunathan et al. (2006) is frequently used in research (e.g., Haasova & Florack, 2019a; Mai & Hoffmann, 2015; Raghunathan et al., 2006), the item wording does not allow for graded responses. Therefore, the first scale aims to capture the unhealthy = tasty belief in a more nuanced way, as participants directly state the extent to which they believe that health causally affects taste and vice versa. To differentiate between the two scales, we will refer to the first scale as causal-belief (CB-Scale) and to the second scale as unhealthy = tasty intuition (UTI-Scale).

To measure the actual health-taste relationship, we adapted measures from previous studies to assess the health-taste correlation in individual meals (Briers et al., 2020; Haasova & Florack, 2019a). We presented 15 pictures of meals, with some meals rather healthy (e.g., Greek salad, couscous) and some rather unhealthy (e.g., meatballs, kebab). Participants indicated their health and taste ratings for each meal on 11-point scales (-5 = *very unhealthy*, +5 = *very healthy*; 1 = *not tasty at all*, 11 = *very tasty*). Health and taste ratings were assessed block-wise, and the order of blocks and images within each block were randomized. We calculated the correlation between health and taste ratings for the 15 meals at the individual level, i.e., we calculated a health-taste correlation coefficient per participant.

To measure dietary intake and overweight, we asked participants to indicate how many serves of fruit and vegetables they eat daily on two 7-point scales (0 = *I don't eat vegetables/fruits*, 6 = *six serves or more*), and we calculated participants' BMI.

4.2. Results

4.2.1. Associations between social class, dietary intake, and the health-taste relationship

Descriptive statistics and Pearson correlations between the measures in this study are reported in Table 2. We see clear indications of dietary inequality in our data. Lower-class individuals ate less fruit ($r = 0.20$, $p < .001$) and vegetables ($r = 0.18$, $p < .001$) and had a higher BMI ($r = -0.13$, $p < .001$), than higher-class individuals. Sixty-three percent of the participants of low social class (1 SD below the mean) were overweight (BMI greater than 25), whereas only 47 percent of participants in the high social class (1 SD above the mean) were overweight, $\chi^2(1, N = 246) = 5.94$, $p = .015$.

Social class was correlated with the CB-Scale of the unhealthy = tasty belief ($r = 0.13$, $p < .001$) and the health-taste correlation ($r = 0.10$, $p < .001$), but not with the UTI-scale of the unhealthy = tasty belief ($r = 0.05$, $p = .109$). Those with a lower subjective sense of standing in society more likely had a negative health-taste correlation and expressed a greater belief that unhealthy food tastes better. All three measures of the relationship between health and taste correlated with daily fruit intake ($r_s = 0.19 - 0.25$, $p_s < 0.001$), daily vegetable intake ($r_s = 0.23 - 0.27$, $p_s < 0.001$), and BMI ($r_s = -0.12 - -0.14$, $p_s < 0.001$). A more negative health-taste relationship (assumed and actual) was associated with an unhealthier dietary intake and a greater likelihood of obesity. Thus, these results stress the importance of the relationship between health and taste for dietary inequality. Further, the two measures of the unhealthy = tasty belief are more strongly correlated with each other ($r = 0.48$, $p < .001$) than with the health-taste correlation ($r_s = 0.35 - 0.42$, $p_s < 0.001$), suggesting that the assumed and actual health-taste relationship are two distinct aspects of the health-taste relationship.

4.2.2. Social class as moderator of the health-taste correlation

The correlation analysis suggests that the health-taste correlation is more negative, the lower individuals' social class. However, the nested structure of health and taste ratings (nested within participants) allows for an analysis of the slope gradients of the health-taste correlation for lower- and higher-class individuals. Specifically, we subjected taste ratings to a linear mixed model with health ratings, social class, and their interaction as fixed effects, and allowed random intercepts for each participant and random slopes for the health ratings using the "lme4" package in R (Bates et al., 2015). The mixed model analyses whether health ratings of each meal predict taste ratings and whether this relationship is moderated by the social class membership of the participants. The results for all fixed and random effects are presented in Table 3.

There was a positive association between health and taste, $b = 0.07$ ($SE = 0.01$), $t(15397) = 6.66$, $p < .001$. Most importantly, the interaction between health ratings and social class was significant, $b = 0.02$ ($SE = 0.01$), $t(15397) = 3.10$, $p = .002$. We then used a spotlight analysis

Table 2

Zero-order correlations for social class and studied variables in Study 1 ($n = 1,027$).

Variables	Mean	SD	1.	2.	3.	4.	5.	6.	7.	8.
1. Social class	5.67	1.70	–							
2. Unhealthy = tasty belief (CB-Scale) ^a	47.66	18.98	0.13**	–						
3. Unhealthy = tasty belief (UTI-Scale) ^a	5.96	2.06	0.05	0.48**	–					
4. Health-taste correlation	0.09	0.43	0.10**	0.35**	0.42**	–				
5. Daily fruit intake	1.67	0.93	0.20**	0.23**	0.19**	0.25**	–			
6. Daily vegetable intake	1.60	0.86	0.18**	0.27**	0.23**	0.24**	0.42**	–		
7. BMI	27.06	6.48	-0.13**	-0.13**	-0.12**	-0.14**	-0.13**	-0.07*	–	
8. Age	51.24	14.35	0.00	0.04	0.04	0.12**	-0.08*	-0.17**	0.22**	–
9. Gender ^b	–	–	0.04	0.11**	0.17**	0.25**	0.11**	0.15**	-0.09**	-0.05

Note. * $p < .05$, ** $p < .01$.

^a High scores on the unhealthy = tasty belief measures (CB-Scale and UTI-Scale) represent a positive health-taste relationship and low scores represent a negative health-taste relationship. Thus, a positive association between social class and the unhealthy = tasty belief means that lower-class individuals have a stronger unhealthy = tasty belief than higher-class individuals.

^b Gender is coded as follows: 0 = men, 1 = women.

Table 3

Parameter estimates of the effects of healthiness and social class on tastiness, using a linear mixed model (LMM) in Study 1.

Predictors	Taste rating Estimates	CI	p
(Intercept)	7.69	7.61 – 7.76	<0.001
Health rating	0.07	0.05 – 0.09	<0.001
Social class	0.02	–0.02 – 0.07	0.369
Health rating × Social class	0.02	0.01 – 0.03	0.002
Random Effects			
σ^2	4.64		
τ_{00} participant	1.28		
τ_{11} participant/health rating	0.08		
ρ_{01} participant	–0.06		
ICC	0.32		
N participant	1,027		
Observations	15,405		
Marginal R^2 / Conditional R^2	0.010 / 0.322		

Note. CI = Confidence interval. σ^2 = residual variance. τ_{00} = random intercept variance. τ_{11} = random slope variance. ρ_{01} = random effect correlation. ICC = intraclass correlation coefficient. The continuous variables in this model, health rating and social class, are centered on the grand mean.

(Aiken & West, 1991) to inspect the health-taste correlation at low (1 *SD* below the mean) and high (1 *SD* above the mean) levels of social class. The analysis revealed that health ratings were significant predictors for expected tastiness only for higher-class individuals, $b = 0.10$ ($SE = 0.03$), $t(1839) = 3.10$, $p = .002$, but not for lower-class individuals, $b = 0.02$ ($SE = 0.03$), $t(1839) = 0.84$, $p = .400$. Fig. 1 illustrates that for higher-class individuals, the healthier they rated the meal, the better the expected taste. Controlling for age, gender, and block order (health or taste ratings measured first) did not affect the results (see Table S1 in the Supplementary Materials).

4.2.3. The health-taste relationship as mediator of social class differences in dietary intake

To test whether the associations between social class, daily fruit and vegetable intake, and between social class and BMI are mediated by the relationship between health and taste (assumed and actual), we calculated mediation analyses using PROCESS in R (model 4) by Hayes (2022). An overview of the indirect effects of social class on fruit and vegetable intake and BMI via the assumed and actual health-taste relationship is depicted in Table 4. We found consistent indirect effects of social class on fruit and vegetable intake and BMI through the health-

Table 4

The Bootstrap 95% confidence interval (Lower and Upper Bounds) for the indirect effect of social class through the relationship between health and taste in Study 1.

Mediators	Predictor variable: Social class					
	DV Fruit intake		DV Vegetable intake		DV Body Mass Index	
	95% CI		95% CI		95% CI	
	Lower	Upper	Lower	Upper	Lower	Upper
Unhealthy = tasty belief (CB-Scale)	0.01	0.05	0.02	0.05	–0.03	–0.01
Unhealthy = tasty belief (UTI-Scale)	–0.00	0.02	–0.00	0.03	–0.01	0.00
Health-taste correlation	0.01	0.04	0.01	0.04	–0.02	–0.01

Note. All variables are standardised. The indirect effect is significant when the confidence interval (CI) does not include zero. Bold font indicates statistical significance. DV: Dependent variables.

taste correlation in individual meals and through the CB-Scale, but not through the UTI-Scale of the unhealthy = tasty belief.

4.3. Discussion

In Study 1, we show that an association between social class and the relationship between health and taste exists and we distinguish between an assumed (unhealthy = tasty belief) and an actual (correlation between health and taste ratings) health-taste relationship. For the assumed health-taste relationship, we find social class differences only on the CB-Scale, but not on the UTI-Scale. On the CB-Scale, lower-class individuals are more likely to express an unhealthy = tasty belief than higher-class individuals. For the actual health-taste relationship, our results suggest that the correlation between health and taste is mainly positive for higher-class individuals. We found no correlation between the healthiness and tastiness of individual meals among lower-class individuals. Furthermore, our results suggest that the relationship between health and taste mediates the association between social class and dietary intake (daily fruit and vegetable intake) and between social class and overweight (BMI). In conclusion, Study 1 provides initial evidence that the relationship between health and taste is an underlying mechanism of dietary inequality between social classes in Germany.

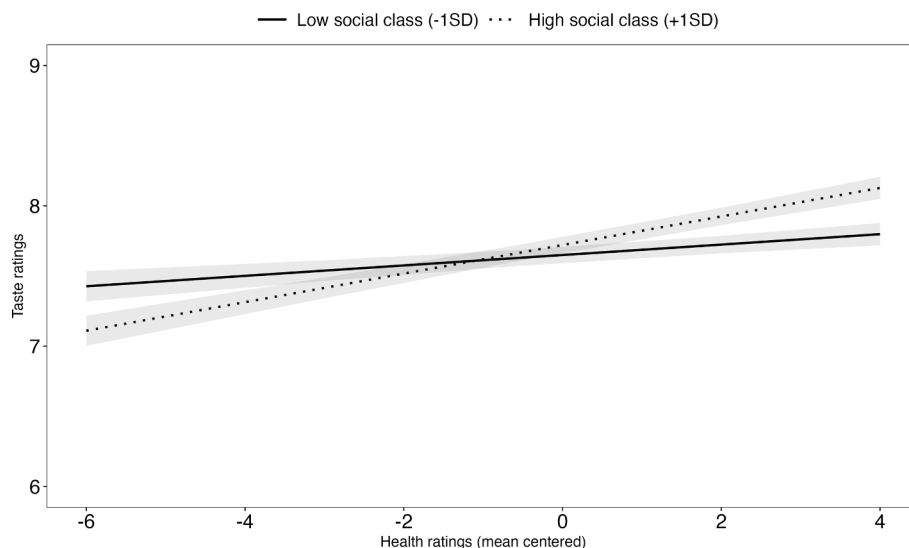


Fig. 1. Line plot depicting the interaction between social class and health ratings on tastiness in Study 1. Note. The interaction effect was created using the “effects” package in R (Fox et al., 2016). Grey areas surrounding the lines indicate the standard errors.

5. Study 2: Longitudinal study in the Netherlands

A limitation of Study 1 is that all variables were measured simultaneously. Therefore, the results of Study 1 may be affected by common method bias (Podsakoff et al., 2003). Study 1 also cannot answer the question of whether social class membership and the health-taste relationship, which existed at an earlier point in time, predict dietary intake and overweight at a later point in time. To strengthen the conclusions drawn about the effects of social class on the relationship between health and taste, dietary intake, and overweight, we analysed longitudinal panel data in Study 2. Specifically, we investigated social class, the assumed health-taste relationship, dietary intake, and overweight at three time points (i.e., 2011, 2016, and 2018) using a subset of a stratified national sample of households in the Netherlands (Longitudinal Internet Studies for the Social Sciences; <https://www.lissdata.nl>). The LISS panel consists of 5,000 households, and each month, sub-samples of these households are contacted to complete a questionnaire with questions relevant to social scientists. One household member provides all the data for the household. As this is a secondary data analysis, we were limited by the variables collected by the panel. Thus, in Study 2, we only analysed the assumed health-taste relationship (i.e., the unhealthy = tasty belief).

5.1. Method

Our data analysis consisted of three waves. Wave 1 included the social class status from 2011 (LISS Economic Situation: Income). Participants selected a rung on the “ladder of life”, with the first rung representing the worst possible life and the tenth rung representing the best possible life. This is a slight modification of Study 1, where we used the instructions provided by Adler et al. (2000) for the social class ladder.

Wave 2 entailed the unhealthy = tasty belief in 2016 (LISS Lifestyle Study), and participants indicated their agreement on a 5-point scale (1 = *disagree entirely*, 5 = *agree entirely*) on five items: “Things that are good for me rarely taste good”, “Eating healthy means sacrificing taste”, “Food that is unhealthy generally tastes better”, “There is usually a trade-off between health and taste in food”, and “Healthy food usually tastes good” (Cronbach’s $\alpha = 0.77$). Compared to Study 1, one item from the UTI-Scale from Raghunathan et al. (2006) was dropped (“There is no way to make food healthier without sacrificing taste”), while three additional items were added to the LISS panel. The UTI-Scale was also changed from a 9-point to a 5-point scale. We recoded items so that high scores indicated a positive assumed health-taste relationship.

In Wave 3, participants reported their fruit and vegetable consumption on a 6-point scale (1 = *never*, 6 = *every day*) and their weight and height in 2018 (LISS Core Health Study). Fruit and vegetable consumption was measured differently than in Study 1. In Study 1, we asked

about daily fruit and vegetable intake. In the LISS panel, participants indicated whether they consumed fruit and vegetables never, a couple of times per month, weekly, multiple times per week, or daily. After matching all three data sets, $N = 794$ participants (48% women, 52% men) with a mean age of 60.75 years ($SD = 14.30$) remained for the final data analysis. The mean BMI was 26.67 ($SD = 7.36$).

5.2. Results

5.2.1. Associations between social class, dietary intake, and the health-taste relationship

Descriptive statistics and Pearson correlations between the measures in this study are reported in Table 5. Again, higher-class individuals consumed more fruit and vegetables ($r_s = 0.12 - 0.17$, $p_s < 0.001$), than lower-class individuals. However, there was no association between social class and BMI ($r = -0.05$, $p = .135$). Social class was correlated with the unhealthy = tasty belief ($r = 0.18$, $p < .001$), and the unhealthy = tasty belief was correlated with fruit and vegetable consumption ($r_s = 0.19 - 0.21$, $p_s < 0.001$) and BMI ($r = -.11$, $p = .003$). Lower-class individuals expressed a stronger belief that unhealthy food tastes better and individuals with a stronger unhealthy = tasty belief ate less fruit and vegetables and had a higher BMI.

5.2.2. The health-taste relationship as mediator of social class differences in dietary intake

We computed mediation analyses using PROCESS in R (model 4) by Hayes (2022) to test whether the unhealthy = tasty belief mediates the associations between social class and fruit and vegetable consumption. We did not perform a mediation analysis with BMI as dependent variable due to the non-significant relationship between social class and BMI. We found consistent indirect effects of social class in 2011 on fruit and vegetable consumption in 2018 through the unhealthy = tasty belief in 2016. The confidence interval of the indirect effects did not include zero for fruit consumption, $b = 0.03$, ($SE = 0.01$), 95% CI [0.01-0.05], and for vegetable consumption $b = 0.03$, ($SE = 0.01$), 95% CI [0.02-0.05]. A summary of the mediation models, including path coefficients, is shown in Fig. 2a and b.

5.3. Discussion

In Study 2, we replicated the findings of Study 1 in a longitudinal design with a Dutch population. We again demonstrate that an association between social class and the health-taste relationship exists, and that the health-taste relationship mediates the association between social class and dietary intake at a later time point. In contrast to Study 1, we did not find a significant association between social class and BMI. Moreover, whereas in Study 1 we did not find an association between social class and the UTI-Scale of the unhealthy = tasty belief, this

Table 5
Zero-order correlations for social class and studied variables in Study 2 ($n = 794$).

Variables	Mean	SD	1.	2.	3.	4.	5.	6.
<i>Wave 1: 2011</i>								
1. Social class	7.25	1.26	–					
<i>Wave 2: 2016</i>								
2. Unhealthy = tasty belief (UTI-Scale) ^a	3.85	0.75	0.18**	–				
<i>Wave 3: 2018</i>								
3. Fruit consumption	4.86	1.35	0.17**	0.19**	–			
4. Vegetable consumption	4.81	1.23	0.12**	0.21**	0.35**	–		
5. BMI	26.67	7.36	-0.05	-0.11**	-0.00	-0.05	–	
6. Age	60.75	14.30	0.06	0.04	0.22**	0.00	0.05	–
7. Gender ^b	–	–	-0.03	0.13**	0.12**	0.12**	0.02	-0.10**

Note. * $p < .05$, ** $p < .01$.

^a High scores on the unhealthy = tasty belief measure (UTI-Scale) represent a positive health-taste relationship and low scores represent a negative health-taste relationship. Thus, a positive association between social class and the unhealthy = tasty belief means that lower-class individuals have a stronger unhealthy = tasty belief than higher-class individuals.

^b Gender is coded as follows: 0 = men, 1 = women.

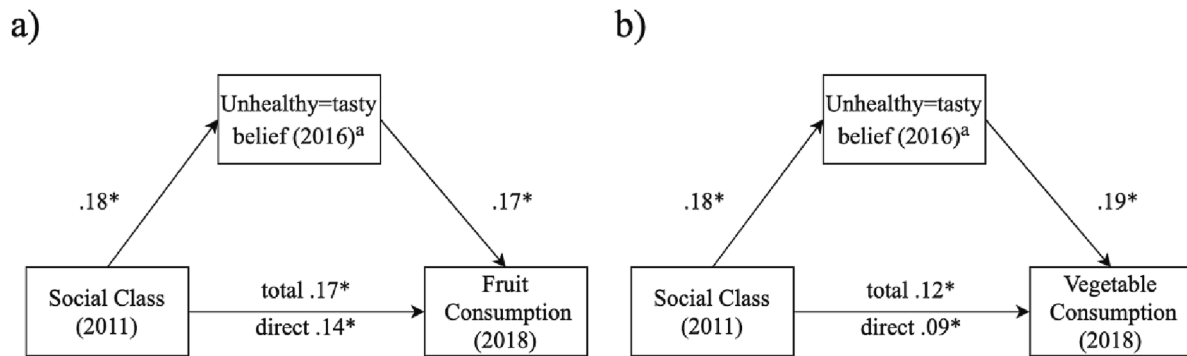


Fig. 2. Summary of the results of mediation analyses with fruit (Fig. 2a) and vegetable (Fig. 2b) consumption as outcome variables in Study 2. Note. * $p < .05$. All variables are standardised. ^a High scores on the unhealthy = tasty belief measure (UTI-Scale) represent a positive health-taste relationship and low scores represent a negative health-taste relationship. Thus, a positive association between social class and the unhealthy = tasty belief means that lower-class individuals have a stronger unhealthy = tasty belief than higher-class individuals.

association was significant in Study 2, and the unhealthy = tasty belief partially mediated the associations between social class and fruit as well as vegetable consumption. Study 2 strengthens our conclusion that social class is related to the health-taste relationship and that this association may help to explain the relationship between social class and dietary intake.

6. General discussion

The reasons for dietary inequality in high-income countries remain unclear, and the underlying psychological mechanisms of the relationship between social class and dietary intake are poorly investigated (Ball et al., 2006; Oude Groeniger et al., 2017). This research demonstrates that the relationship between health and taste contributes to social class differences in dietary intake. In a large-scale cross-sectional data set of a German population and in a longitudinal study of a Dutch sample, we found largely consistent support that social class is associated with the health-taste relationship (assumed and actual) and that the association between social class and dietary intake is mediated by the health-taste relationship.

6.1. Social class and the health-taste relationship

Our findings are in line with research that food beliefs are dependent on the socioeconomic position of an individual in society, and that these food beliefs may contribute to the frequently reported dietary inequality between social classes (van der Heijden et al., 2021). To the best of our knowledge, this is the first study to find social class differences in the health-taste relationship. In addition, our research differentiates between an assumed and actual health-taste relationship. The assumed relationship is reflected in consumers' belief whether unhealthy food is in general tastier than healthy food, whereas the actual relationship is reflected in the correlation between consumers' health and taste ratings of individual meals. While the measures for the unhealthy = tasty belief correlate with the health-taste correlation at levels that are consistent with medium effect sizes, these correlations are far from perfect. This indicates that the assumed and actual health-taste relationship are relatively independent aspects of the health-taste relationship.

Regarding the assumed health-taste relationship, our results support research suggesting that the unhealthy = tasty belief is dependent on the social context (Briers et al., 2020; Werle et al., 2013). While Briers et al. (2020) investigated the unhealthy = tasty belief across countries, we examined social class as cultural construct (Kraus et al., 2011), extending this line of research.

In Study 1, we observed an association between social class and the unhealthy = tasty belief measured with the CB-Scale, but not when using the UTI-Scale with the items from Raghunathan et al. (2006) and Mai

and Hoffmann (2015). Interestingly, the association between social class and the UTI-Scale was significant in Study 2 with a Dutch population. An important difference between the two studies is that the UTI-Scale in Study 2, like the CB-Scale in Study 1, allowed participants to express their unhealthy = tasty belief in a more nuanced way than the original items of the UTI-Scale by Raghunathan et al. (2006) applied in Study 1. In Study 2, we used secondary data from a Dutch panel where the unhealthy = tasty belief was assessed using a five-item Likert-scale. The panel replaced one item from the original scale with three more moderately worded items.

One explanation for our finding that the assumed health-taste relationship is associated with social class may be that lower-class individuals grow up and live in unhealthier food environments than higher-class individuals (Burgoine et al., 2017; Drewnowski et al., 2013) and thus, may learn that unhealthy food tastes better just because healthy food is not well represented. Indeed, recent research demonstrated that food environments with many unhealthy and tasty foods can lead to negative perceptions of the health-taste relationship, regardless of the actual correlation between health and taste in the food environment (Kunz et al., 2023).

Regarding the actual health-taste relationship, previous research and the results of our study indicate that, on average, there is a positive health-taste correlation (Briers et al., 2020; Haasova & Florack, 2019a). However, our research suggests that this positive relationship is stronger for higher-class than lower-class individuals. In fact, for lower-class individuals, we found no correlation at all between health and taste ratings. This implies that higher-class individuals tend to infer a good taste as well as health of a food item from the same cues. On the other hand, lower-class individuals seem to use different cues when judging the taste than when judging the health of food. A possible explanation for the differential use of cues for forming taste and health expectations could be that lower-class individuals are less often exposed to cues that signal both health and taste, like fresh ingredients in meals. Additionally, higher-class individuals may respond to food cues differently by recalling prior taste experiences that lower-class individuals might not have.

6.2. Health-taste relationship as mediator of social class differences in dietary intake

The assumed and actual health-taste relationship partially mediated the association between social class and the dietary intake in both studies, and the assumed health-taste relationship mediated the association between social class and BMI in Study 1. Taken together, these findings suggest that lower-class individuals who have a more negative health-taste relationship (assumed and actual) are more likely to consume unhealthy foods, leading to an increased risk of overweight and

obesity.

While material resources and the food environment have been considered as underlying mechanisms of dietary inequality (Burgoine et al., 2017; Rao et al., 2013; Stormacq et al., 2019), our research proposes the health-taste relationship as an additional mechanism that has been largely overlooked. In one study, Oude Groeniger et al. (2017) reported that the higher consumption of so-called “superfoods” (products with high amounts of antioxidants, vitamins, and minerals) by higher-class individuals is a consequence of social distinction efforts. We add to this literature by highlighting that a socially patterned relationship between health and taste lead to differential consumption of fruits and vegetables and increased obesity risk in lower-class individuals.

6.3. Practical implications

Our finding that the health-taste relationship (assumed and actual) partially explains the association between social class and dietary intake has practical implications. While it is necessary to make healthy food options more affordable and accessible to lower-class individuals, policy makers should also focus on changing the health-taste relationship among lower-class individuals. Policymakers often focus on cognitive approaches when combating the obesity pandemic by providing nutritional information and labelling foods as healthy or unhealthy (for a meta-analysis, see Ikonen et al., 2020). Our study results question whether these strategies achieve the same effects for lower- and higher-class individuals. For higher-class individuals with a positive health-taste relationship, health labels and nutritional information may facilitate the selection of healthy products that are perceived as tasty. However, for lower-class individuals with a negative health-taste relationship, health labels may be counterproductive, as they may be avoided due to low taste expectations.

Instead, policy makers should try to change people's lay beliefs about the relationship between health and taste, for example by creating food environments with many healthy and tasty food options (see also Kunz et al., 2023). One difficulty is that food beliefs develop in childhood and can persist into adulthood (Puhl & Schwartz, 2003). Changing the public food environments which children are exposed to, such as school cafeterias, may therefore be an effective intervention. Ideally, changing the assumed relationship between health and taste may also affect the actual relationship between health and taste, as lower-class individuals may be more inclined to try healthier foods and learn that the same cues can signal health as well as good taste.

6.4. Limitations

This research does not come without limitations. One limitation of our research is that the correlations between social class and the relationship between health and taste in this report are small. This limits the interpretability of our findings, but it should be noted that much higher correlations could not be expected. Social class is a broad concept that covers and influences many aspects of people's lives, whereas the relationship between health and taste is a domain-specific construct. Albeit our correlations are small, they are similar to other correlations between social class and psychological constructs reported in the literature (e.g., Belmi et al., 2020). Nevertheless, it is beyond doubt that there is a link between social class and obesity, and that it comes with major health consequences. Small correlations might be considered negligible for single events but have considerable implications over a longer period of time (Funder & Ozer, 2019). Thus, the associations between social class and the health-taste relationship may not be predictive for single food choices, but may have serious implications for multiple food choices.

Another limitation of our research is that we cannot establish a causal relationship between our variables. In Study 1, we analysed cross-sectional data and cannot rule out the possibility that fruit and vegetable intake influences the health-taste relationship and not the other way around. To draw stronger conclusions, we introduced temporal order

between our variables in Study 2. As a result, we could conclude that the unhealthy = tasty belief in 2016 relates to fruit and vegetable consumption in 2018 and that both, the unhealthy = tasty belief and dietary intake, are associated with social class in 2011. The longitudinal design also reduced common method bias present in Study 1 (Podsakoff et al., 2003).

6.5. Directions for future research

Future research should investigate the formation of the assumed and actual health-taste relationship, and how they relate to social class. The assumed health-taste relationship (unhealthy = tasty belief) may be learned by children at an early age and may depend on parental styles and the composition of the food environment (Damen et al., 2019; Kunz et al., 2023). For the actual health-taste relationship, we argue that the direction and strength of the relationship depends on cue recognition and utilization, and that a negative correlation may stem from the use of different cues or different use of the same cues to assess the health and taste of food options. Future research should investigate how lower-class and higher-class individuals recognise and exploit cues when evaluating the healthiness and tastiness of a food item. Recently, many visual and cognitive cues have been accumulated in the literature that signal a food product's health (Chan & Zhang, 2022). However, it is unclear whether the same cues also signal tastiness and whether this is the case for everyone.

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8. Ethics statements

Study 1 of this research was conducted in accordance with the Declaration of Helsinki (revised 1983) and local guidelines of the Faculty of Psychology, University of Vienna. Participants in Study 1 were informed about its aim and confidentiality of the data collection, and gave their consent to participate. Participants could also withdraw at any time. The study was approved by the Institutional Review Board (IRB) of the department for occupational, economic and social psychology of the University of Vienna (Approval number: 2021_S_01A). We did not obtain ethical approval for Study 2 because the data is from a secondary source but ethics approvals are not required by law in the Netherlands for questionnaire research among adults (<https://www.lissdata.nl/faq-page#n5512>).

CRediT authorship contribution statement

Niklas Pivecka: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Visualization. **Sonja Kunz:** Conceptualization, Methodology, Validation, Writing – review & editing. **Arnd Florack:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodqual.2023.104914>.

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